

# Chapter 8:

# Integration Techniques and Improper Integrals

Section 8.2: Integration by Parts

## Integration by Parts: Indefinite Integrals

The Product Rule states that if $f$ and $g$ are differentiable functions, then

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

In the notation for indefinite integrals this equation becomes

$$\int [f(x)g'(x) + g(x)f'(x)]\,dx = f(x)g(x)$$

or

$$\int f(x)g'(x)\,dx + \int g(x)f'(x)\,dx = f(x)g(x)$$

We can rearrange this equation as

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$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

Formula 1 is called the **formula for integration by parts**.

**1**

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

Formula 1 is called the **formula for integration by parts**. It is perhaps easier to remember in the following notation. Let $u = f(x)$ and $v = g(x)$. Then the differentials are $du = f'(x)\,dx$ and $dv = g'(x)\,dx$, so, by the Substitution Rule, the formula for integration by parts becomes

$$\int u\,dv = uv - \int v\,du$$

## EXAMPLE

Find $\int x \sin x \, dx$.

It is helpful to use the pattern:

$u = \square$ $\quad dv = \square$

$du = \square$ $\quad v = \square$

**SOLUTION**

$$I = -x\cos x + \sin x + C$$

## EXAMPLE

Find $\int x \sin x \, dx$.

**NOTE** Our aim in using integration by parts is to obtain a simpler integral than the one we started with. Thus in Example 1 we started with $\int x \sin x \, dx$ and expressed it in terms of the simpler integral $\int \cos x \, dx$. If we had instead chosen $u = \sin x$ and $dv = x \, dx$, then $du = \cos x \, dx$ and $v = x^2/2$, so integration by parts gives

$$\int x \sin x \, dx = (\sin x) \frac{x^2}{2} - \frac{1}{2} \int x^2 \cos x \, dx$$

Although this is true, $\int x^2 \cos x \, dx$ is a more difficult integral than the one we started with. In general, when deciding on a choice for $u$ and $dv$, we usually try to choose $u = f(x)$ to be a function that becomes simpler when differentiated (or at least not more complicated) as long as $dv = g'(x) \, dx$ can be readily integrated to give $v$.

## EXAMPLE

Evaluate $\int \ln x \, dx$.

It is helpful to use the pattern:

$u = \square$ $\quad dv = \square$

$du = \square$ $\quad v = \square$

**SOLUTION**

$$I = x \ln x - x + C$$

## EXAMPLE

Find $\int x^2 e^x \, dx$

It is helpful to use the pattern:

$u = \square$ $\quad dv = \square$

$du = \square$ $\quad v = \square$

## SOLUTION

$$I = x^2 e^x - 2xe^x + 2e^x + C$$

## EXAMPLE

Find $\int e^x \sin 2x \ dx$.

It is helpful to use the pattern:

$u = \square$ $\quad dv = \square$

$du = \square$ $\quad v = \square$

## SOLUTION

$$I = -\frac{2}{5}e^{x}\cos(2x) + \frac{1}{5}e^{x}\sin(2x) + C$$

## EXAMPLE

Find $\int \sec^3 x \, dx$.

It is helpful to use the pattern:

$u = \square$ $\quad dv = \square$

$du = \square$ $\quad v = \square$

**SOLUTION**

$$I = \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$$

## EXAMPLE

Find $\int \arcsin y \; dy$.

It is helpful to use the pattern:

$u = \square$ $\quad dv = \square$

$du = \square$ $\quad v = \square$

**SOLUTION**

$$I = y \arcsin y + \sqrt{1 - y^2} + C$$

## Integration by Parts: Definite Integrals

If we combine the formula for integration by parts with Part 2 of the Fundamental Theorem of Calculus, we can evaluate definite integrals by parts. Evaluating both sides of Formula 1 between $a$ and $b$, assuming $f'$ and $g'$ are continuous, and using the Fundamental Theorem, we obtain

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$$\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big]_a^b - \int_a^b g(x)f'(x)\,dx$$

## EXAMPLE

Find $\int_1^2 x^2 \ln x \ dx$

**SOLUTION**

$$I = \frac{8}{3}\ln(2) - \frac{7}{9}$$

**REMARK** You can use the acronym LIATE as a guideline for choosing $u$ in integration by parts. In order, check the integrand for the following.

Is there a Logarithmic part?

Is there an Inverse trigonometric part?

Is there an Algebraic part?

Is there a Trigonometric part?

Is there an Exponential part?

## SUMMARY: COMMON INTEGRALS USING INTEGRATION BY PARTS

**1.** For integrals of the form

$$\int x^n e^{ax}\,dx, \quad \int x^n \sin ax\,dx, \quad \text{or} \quad \int x^n \cos ax\,dx$$

let $u = x^n$ and let $dv = e^{ax}\,dx$, $\sin ax\,dx$, or $\cos ax\,dx$.

**2.** For integrals of the form

$$\int x^n \ln x\,dx, \quad \int x^n \arcsin ax\,dx, \quad \text{or} \quad \int x^n \arctan ax\,dx$$

let $u = \ln x$, $\arcsin ax$, or $\arctan ax$ and let $dv = x^n\,dx$.

**3.** For integrals of the form

$$\int e^{ax} \sin bx\,dx \quad \text{or} \quad \int e^{ax} \cos bx\,dx$$

let $u = \sin bx$ or $\cos bx$ and let $dv = e^{ax}\,dx$.

## EXAMPLE Using the Tabular Method

Pay Attention
This is Important

Find $\int x^3 e^{2x} dx$

**SOLUTION** Let $u = x^3$ and $dv = v'dx = e^{2x}dx$ then create a table as follow:

Alternate signs starting from positive

| $x$ and its derivatives | | $v'$ and its antiderivatives |
|---|---|---|
| $x^3$ | + | $e^{2x}$ |
| $3x^2$ | − | $\frac{1}{2}e^{2x}$ |
| $6x$ | + | $\frac{1}{4}e^{2x}$ |
| $6$ | − | $\frac{1}{8}e^{2x}$ |
| $0$ | | $\frac{1}{16}e^{2x}$ |

$$\int x^3 e^{2x} dx = \frac{1}{2}x^3 e^{2x} - \frac{3}{4}x^2 e^{2x} + \frac{3}{4}x e^{2x} - \frac{3}{8}e^{2x} + C.$$

$$I = \frac{1}{2}x^3e^{2x} - \frac{3}{4}x^2e^{2x} + \frac{3}{4}xe^{2x} + \frac{3}{8}e^{2x} + C$$

EXAMPLE

**Using Two Methods Together** **In Exercises 63–66, find the indefinite integral by using substitution followed by integration by parts.**

**63.** $\int \sin\sqrt{x}\, dx$

SOLUTION

$$I = 2\sin\sqrt{x} - 2\sqrt{x}\cos\sqrt{x} + C$$

## EXAMPLE

Find $\int x \cos(x^2) \ dx.$

$$I = \frac{1}{2}\sin x^2 + C$$

Solve exercises of section 8.2

Start with the suggested problems mentioned in the syllabus.